Steven Cleasby-Mayeda

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EDUCATION

Oregon State University *Corvallis, OR*
M.S. in Artificial Intelligence GPA: 3.8 *Expected June 2027*

- **Relevant Coursework:** Foundation Models, Deep Learning, Optimization for AI, Data Cleaning for ML, Operating Systems II, Graph Theory

Oregon State University *Corvallis, OR*
B.S. Applied Computer Science (AI Focus) GPA: 3.37, Major GPA: 3.52 *June 2026*

- Honor Roll (7 terms)
- **Relevant Coursework:** Web Development, Databases, Software Engineering, Algorithms, Machine Learning & Data Mining, Mathematical Statistics, Parallel Programming, Reverse Engineering Malware

EXPERIENCE

Waste Treatment Completion Company *Richland, WA. (Remote)*
Leading Software Engineer (Senior Capstone Project) *Sep 2025 – June 2026*

Project Page

- Led a team of developers building a Python data-analysis application for safety-critical radioactive waste vitrification calculations at the Hanford site.
- Architected validated calculation pipelines with an emphasis on auditability, numerical consistency, and long-term maintainability.
- Managed task delegation, code reviews, and interpersonal conflict resolution across the student engineering team, and authored a developer handover package ensuring continuity for future maintainers.

Forever Sabah *Sabah, Malaysia*
Research Intern *Jun 2023 – Sep 2023*

- Built data processing scripts for climate raster datasets, analyzing precipitation outlier patterns using QGIS.
- Designed and implemented database schemas for wildlife image datasets supporting individual elephant re-identification.

RESEARCH & PROJECTS

Quantitative ML for Systematic Trading | Independent Research *Python, PyTorch, LightGBM*

- Built an OSU high performance compute research pipeline with experiment tracking and leakage-safe evaluation.
- Trained a Gradient boost model realized-volatility forecaster beating a HAR-RV baseline out-of-sample.
- Designed a post-earnings-drift signal on a 40-year, 446K-event WRDS panel, clearing all pre-registered gates: NW $t = 6.18$, permutation $p = .005$, net Sharpe 1.10@2bp.

ESM3 Representations for HIV bnAb Modeling | Course Project *PyTorch, ESM3, LoRA, scikit-learn*

Project Page

- Built an end-to-end ML pipeline utilizing frozen and LoRA-adapted ESM3 (1.4B param) representations for HIV bnAb breadth prediction on CATNAP (AUROC improved from 0.651 to 0.773).

- Engineered a grouped lineage-held-out cross-validation framework with bootstrap uncertainty estimation to prevent data leakage across biologically related antibodies.
- Trained a categorical VAE on whole heavy-chain sequences for latent-space lineage interpolation, recovering known VRC34 intermediates in 2 of 4 lineage triples.

Multi-Agent Orchestration Harness *Python, Git, Agentic Systems*

- Architected a git-worktree-isolated harness in which an orchestrator agent delegates coding/research tasks to a crew of parallel LLM agents (Claude, Codex, Grok).
- Built usage-based rate-limit pacing/routing across provider windows and an A/B testing framework to statistically compare agent protocols.

Parallel Logic Expert with Operator Routing | Course Project *Python, PyTorch*

Project Page

- Designed and implemented a modular mixture-of-experts (MoE) transformer featuring differentiable fuzzy-set logical operators, learned operator routing, and backbone cross-attention in PyTorch.
- Built a configurable training and evaluation pipeline supporting ablation studies and routing diagnostics, achieving empirical performance gains on logical reasoning benchmarks.

Guidance – Agentic Field Technician Assistant | 1st Place, BeaverHacks 2026 *Gemini, Python*

GitHub Repository | Demo Video | Project Submission

- Won 1st place (beginner track, 20 teams) at BeaverHacks 2026, architecting and shipping the full-stack solution in 24 hours.
- Developed a multimodal Gemini agent pipeline integrating image/audio input, transcription, local RAG, function calling, and image editing with QR-scoped retrieval.

Backend Web Application / Personal Website *JavaScript, React, REST APIs*

- Built a MERN-stack application with authentication, database integration, and REST-style backend endpoints.

TECHNICAL SKILLS

Languages: Python, C, JavaScript, SQL

Web & Backend: React/MERN, REST APIs, authentication, relational database design

Machine Learning: PyTorch, HuggingFace Transformers, LightGBM/gradient boosting, transformer design, training/evaluation, agentic systems, RAG, scikit-learn

Data & Evaluation: NumPy, pandas, statistics, feature engineering, walk-forward/purged cross-validation, backtesting, Weights & Biases

Systems & Tools: Linux, CUDA, SLURM/HPC, Docker, Git/GitHub, cloud infrastructure (AWS), data structures & algorithms

Practices: Testing, CI/CD, code review, technical documentation, Agile development